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EDUCATION

Islamic University of Technology (IUT)

B.Sc. in Electrical and Electronic Engineering (4th Year, CGPA: 3.64/4.00)

Gazipur, Bangladesh

2023 – 2027 (Expected)

Govt. Edward College, Pabna | *Higher Secondary Certificate (Science) — GPA 5.00/5.00*

2022

Pabna Zilla School, Pabna | *Secondary School Certificate (Science) — GPA 5.00/5.00*

2020

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Transportation Engineering

Gazipur, Bangladesh

Supervisor: Dr. Nazmus Sakib, PhD, PE — Dept. of Civil & Environmental Engineering, IUT

2025 – Present

- Design and build embedded instrumentation and data-acquisition systems for transportation research spanning vehicle braking dynamics, fleet tracking, roadside air quality and laboratory material testing.
- Instrumented a cycle rickshaw with encoder-based speed/distance measurement and time-of-flight brake-event detection, producing RTC-timestamped per-run datasets for braking-performance analysis.
- Developed laboratory readout tooling — multi-port load-cell acquisition and digital dial-indicator decoding — replacing manual reading with time-aligned CSV capture; also deployed a GPS-tagged particulate-matter logger and an RFID/QR rickshaw fleet management system.

Optical IRS for Visible Light Communication (Li-Fi)

Gazipur, Bangladesh

Lead & Corresponding Author — Independent Simulation Research

2026 – Present

- Built an end-to-end Python channel-simulation framework for mirror-array optical intelligent reflecting surfaces (OIRS) in VLC, covering geometry, diffuse reflection, mobility, blockage and receiver noise.
- Identified two incompatible mirror reflection models in the OIRS-VLC literature and showed that omitting the law of reflection **overstates SNR by 6.3 dB**, with the error growing as the array scales.
- Derived a closed-form bisector orientation update that beats metaheuristic mirror-steering search by **five orders of magnitude** in compute, and showed control latency (13.0 dB) dominates blockage (1.1 dB) as a loss driver.
- Released a fully reproducible codebase — 12 publication figures, BER and power sweeps, and optimizer validation — at github.com/thisisdibbo/oirs-vlc-reflection-model; manuscript in preparation for journal submission.

Biodegradable / Transient Chips with On-Chip AI Integration

Gazipur, Bangladesh

Sole Author — Independent Simulation Research

2026 – Present

- Designing a simulation-only study of biodegradable transient semiconductor devices coupled with on-chip AI inference, targeting a Q1 journal.
- Device-to-circuit workflow spanning **Silvaco TCAD** process/device simulation and **Cadence Virtuoso** circuit design, with Python and MATLAB for extraction and analysis.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, MATLAB, Arduino C/C++, Kotlin (Jetpack Compose), $\text{\LaTeX}$

Simulation & CAD: MATLAB, Proteus, PSpice, COMSOL, PSIM, LINKPLANNER, Tinkercad, Silvaco TCAD, Cadence Virtuoso, Lumerical FDTD, SolidWorks, CYME PSAF, EasyEDA, EAGLE PCB

Hardware & Systems: ESP32, Arduino (Uno / Nano / Mega), Circuit Design, PCB Layout & Fabrication, Power Electronics, MOSFET & H-Bridge Motor Drivers, FPV Drone Build & Tuning, 3D Printing

Sensors & Interfaces: GPS (NEO-6M / M8N), IMU (MPU6050), MAX30102, MLX90614, DHT22, PMS5003, VL53L0X ToF, MQ Gas, TDS / DS18B20, RFID, TFT & OLED, I²C / SPI / UART / 1-Wire / ESP-NOW

Software & Data: PyQt6 / PySide6 Desktop Apps, Firebase Realtime Database, SQLite, SD / CSV Data Logging, Git & GitHub, PLC Ladder Logic (LS Electric XG5000), Betaflight, EdgeTX, GRBL

Languages: Bangla (Native), English (Professional Working Proficiency)

ACADEMIC & ENGINEERING PROJECTS

Smart Audio Recognition System | Contributor — Python, PyQt5, DSP, DTW, SQLite

2026

- Multi-modal music information retrieval engine that identifies original tracks, cover versions, instrumental renditions and hummed queries from live microphone input — 98% accuracy on clean audio, 67.5% on humming.
- Combines spectral-peak combinatorial hashing, PCEN-filtered chromagrams with dynamic time warping, and probabilistic YIN pitch tracking, behind a multithreaded PyQt5 interface over SQLite.

Smart Medical Box | ESP32, ESP-NOW, TFT_eSPI, 3D Printing

2025 – 2026

- Automated medicine cabinet: three motorised pill silos and two peristaltic syrup pumps, with a piezo disc under the tray so the box counts tablets it has *actually* delivered rather than ones it assumes.
- Integrated MAX30102, MLX90614 and DHT22 patient vitals behind a 480×320 TFT interface, over a master-slave ESP-NOW link, in a 3D-printed enclosure.

WaterSCADA — Water Quality Monitoring System | ESP32, PyQt6, GPS, SCADA

2025 – 2026

- GPS- and RTC-stamped logger measuring TDS/EC and DS18B20 water temperature, appending every reading to microSD CSV and streaming it over USB serial.
- Built a PyQt6 SCADA dashboard — animated gauges, dark GPS track map, rolling trend charts, one-click CSV export — packaged as a single Windows executable.

Portable Air Quality Monitoring System | ESP32, PMS5003, GPS, RTC, Firebase

2025

- Portable particulate-matter logger: PMS5003 PM1.0/2.5/10 readings tagged with NEO-6M GPS position and DS3231 RTC time, logged to SD card in CSV.
 - Earlier revision streamed live readings to a Firebase-backed web dashboard for remote viewing and record download from anywhere.
- RFID Rickshaw Management System** | *Python, Kotlin, Firebase, RFID* 2025
- Master/slave RFID pairing: the owner scans a master card, the puller scans a slave card, and the puller is assigned to that rickshaw in a single transaction.
 - Delivered three linked clients — a Python desktop data-entry app, a Kotlin owner mobile app and a public QR lookup page — synchronised through Firebase Realtime Database.
- Custom IBT-2 (BTS7960) H-Bridge Motor Driver PCB** | *PCB Design, Power Electronics* 2025
- Designed, fabricated and bench-tested a custom high-current H-bridge driver board; measured output voltage equal to input voltage under load.
 - Added an on-board HAT circuit — power LED, two signal LEDs and a JST header — for direct connection to an Arduino shield PCB.
- BreakProject — Braking Distance & Motion Logger** | *ESP32, VL53L0X, Rotary Encoder* 2026
- Measures wheel speed, acceleration and travelled distance from a rotary encoder and detects brake events with a VL53L0X time-of-flight sensor, writing RTC-timestamped CSV logs per test session.
 - Field-tested on a cycle rickshaw for real-world braking-performance measurement.
- Additional Open-Source Work** | *github.com/thisisdibbo* 2025 – 2026
- **Multi Serial Monitor** — PySide6 dashboard reading several QLB9 industrial load-cell indicators over simultaneous COM ports with time-aligned CSV logging.
 - **DialIndicatorReader** — Arduino Nano firmware and PC tooling that decode the clock/data protocol of low-cost digital calipers and dial indicators.
 - **Portable Display** — 3D-designed and printed enclosure for a DIY portable monitor, released with STL files and renders.
- Digital Snake & Ladder Game** | *Digital Logic Design, 74xx ICs, Proteus, PCB Fabrication* 2024 – 2025
- Decomposed a two-player board game into eight hardware modules — dice, player selector, player/snake/ladder position decoders, memory, winner detection, reset and board display.
 - Designed and simulated every module from 74xx combinational and sequential logic in Proteus, integrated them into a full-game schematic, and generated complete Gerber / CAD-CAM fabrication output.
- FPV Drone Builds — Digital & Analog Video Systems** | *ELRS, Betaflight, EdgeTX, RF* 2025 – 2026
- Built and tuned a quadcopter around a digital HD FPV camera and VTX link with an ExpressLRS control link; configured PID tuning, rates, failsafe and OSD, with EdgeTX setup on a RadioMaster TX16S.
 - Built a second airframe on a 5.8 GHz analog camera/VTX chain — power distribution, ESC and motor wiring, antenna selection and full soldering assembly — then compared link latency and robustness in flight testing.
- Laser Engraver with Custom Power Control Panel** | *Power Electronics, GRBL, 3D Printing* 2025 – 2026
- Designed and fabricated an in-house power control panel board for a diode laser engraver, covering mains switching, regulated driver supply and protection circuitry.
 - Modelled and 3D-printed the machine casing and panel enclosure; drives the machine through GRBL firmware with LaserGRBL toolpath control.

INDUSTRIAL TRAINING

- Industrial Attachment Program** Islamic University of Technology
Industrial Trainee Sep. 2026 – Oct. 2026
- **Ghorashal Power Station (Oct. 3 – Oct. 8):** Completed coursework and field modules on thermal power generation, generator synchronization, step-up transformers, and high-voltage grid integration.
 - **Bangladesh Atomic Energy Commission (Sep. 20 – Sep. 24):** Attended 5 days of training on nuclear reactor physics, radiation safety protocols, and advanced instrumentation systems.
 - **Industrial Visits & Seminars (Sep. 14 – Sep. 25):** Evaluated cellular network architecture at Robi Axiata, observed automated PCB fabrication and floating solar PV systems at Walton Hi-Tech, and attended a 5-hour VLSI design seminar by Neural Semiconductor Ltd.

ACHIEVEMENTS & AWARDS

- International Mars Rover Competitions** | *Electrical Sub-Team Member, Project Altair* 2025 – 2026
- **University Rover Challenge (URC):** 2025 Finalist — placed **18th of 36** international teams in the world's premier Mars rover competition; 2026 On-Site Finalist.
 - **International Rover Challenge (IRC):** on-site finalist in both the 2025 and 2026 editions. **Anatolian Rover Challenge (ARC):** 2026 on-site finalist, in autonomy, science and equipment-servicing missions.
- Certifications & Competitions** 2023 – 2026
- IUT AUS Robo Race; PhysicsBrawl Online (Physics Olympiad); certificates of participation in various robotics and technology workshops.

LEADERSHIP & ACTIVITIES

- IEEE Robotics and Automation Society, IUT Student Branch Chapter** | *Joint Secretary* 2026 – Present
- **Instructor, Arduino Workshop (2026):** conducted a hands-on session for **50+ participants**, covering microcontroller fundamentals, sensor interfacing and embedded programming.
 - Coordinate chapter operations, technical workshops and member activities across the robotics and automation community at IUT.
- IUT Robotics Society** | *Project Coordinator* 2023 – Present
- Manage technical workflows, competition preparation and hardware logistics across the society's research and competition teams.
- Project Altair, IUT** | *Senior Electrical Sub-Team Member* 2024 – Present
- Lead power distribution and protection design, embedded control and telemetry integration, PCB design and hardware testing for the team's competition Mars rovers.